Biometric Attendance Management Platform

UX/UI User Flow Specification

Professional user-flow specification for UI/UX design and prototyping

Based on the supplied Product Requirements Document

Biometric Attendance Management Platform

UX/UI User Flow Specification

1. Product Structure

```
The platform should be designed around 7 major areas:
```

1. Authentication

2. Dashboard

3. Attendance

4. Employees

5. Devices

6. Locations / Branches

7. Administration & System Monitoring

```
Primary navigation:
```

```
Dashboard → Overview
```

```
Attendance → Live Attendance → Attendance History → Reports / Analytics
```

```
Employees → Employee Directory → Employee Profile → Biometric Status
```

```
Devices → Device Registry → Device Details → Provision / Activate Device → Device Monitoring
→ Firmware / Remote Actions
```

```
Locations → Locations / Branches → Location Details
```

```
Administration → Users & Roles → Alerts → Audit Log → Integration / Sync → System Settings
```

2. Global Application Flow

OPEN PLATFORM

```
↓
Login
↓
Enter Email + Password
↓
Credentials Valid?
■■■ NO → Error
■■■ YES → MFA Required?
■■■ NO → Dashboard
■■■ YES → Enter MFA → MFA Valid?
■■■ NO → Error
■■■ YES → Dashboard
```

3. First-Time User / Account Setup

Admin Login

```
↓
First-Time Setup
↓
Verify Account
↓
Set Password
↓
Configure MFA
```

```
↓
Review Organization
↓
Dashboard
```

• Account Verification

• Create Password

• MFA Setup

• Recovery Codes

• Organization Overview

• Dashboard

```
Screens:
```

1. Account → Password

2. Security → MFA

3. Organization → Confirm organization information

4. Infrastructure → Add/verify locations

5. Devices → Provision devices

6. Employees → Verify employee synchronization

7. Ready → Enter dashboard

```
Progressive setup:
```

4. Dashboard Flow

Dashboard

```
■
■■■ Attendance Overview
■■■ Live Activity
■■■ Device Health
■■■ Employee Sync
■■■ Alerts
■■■ Location Overview
```

• Attendance Summary: Present, Absent, Late, Clocked In, Clocked Out, Total Attendance Events

• Live Attendance Feed

• Device Health

• Synchronization Status

• Alerts

```
Dashboard sections:
```

5. Attendance Flow

Attendance

```
↓
Live Attendance
↓
Real-Time Attendance Feed
```

• Location

• Device

• Employee

• Direction

• Date

• Status

```
Filters:
```

```
Attendance event:
Employee / Staff Number / Location / Device / Direction / Timestamp / Match Score / Sync Status
```

6. Attendance History Flow

Attendance

```
↓
Attendance History
↓
Apply Filters
↓
Results
↓
Select Attendance Event
↓
Attendance Detail
```

• Employee

• Staff number

• Location

• Device

• In/Out

• Captured timestamp

• Timezone

• Match score

• Event ID

• Sync status

• Web software delivery status

• Retry status if applicable

```
Attendance Detail:
```

```
Failed Delivery:
Attendance captured
↓
Upload to web software
↓
Failed
↓
Retry automatically
↓
Success
```

```
If repeatedly rejected:
Retry → Retry → Retry → Dead Letter Queue → Admin Alert
```

7. Attendance Reports / Analytics

Attendance

```
↓
Reports & Analytics
↓
Select Location
↓
Select Date Range
↓
Select Employee / Department
↓
Generate
↓
Analytics Dashboard
```

• Daily attendance

• Weekly attendance

• Monthly attendance

• Employee attendance

• Location attendance

• Late arrivals

• Absences

• Attendance trends

```
Report views:
```

```
Export:
Report → Export → CSV / Excel / PDF
```

8. Employee Directory Flow

Employees

```
↓
Employee Directory
↓
Search / Filter
↓
Employee
↓
Employee Profile
```

• Name

• Staff Number

• Location

• Employment Status

• Biometric Status

• Last Sync

• Attendance Status

```
Employee Directory:
```

```
Important:
Employees are not created manually inside the biometric platform. The existing web software is
the source of truth for employee records, assignments and employment status.
```

9. Employee Profile Flow

Employee Directory

```
↓
Select Employee
↓
Employee Profile
```

```
Profile sections:
```

• Full name

• Staff number

• Web reference ID

• Employment status

```
Identity:
```

• Assigned location

• Branch

```
Assignment:
```

• Enrolment status

• Number of enrolled fingers

• Last biometric sync

• Enrolled device/location

```
Biometric:
```

• Today's status

• Recent attendance

• Attendance history

```
Attendance:
```

```
Restriction:
There should be no "Create Employee" button in this system.
```

Employee missing?

```
↓
"Sync from Web Software"
↓
Employee appears after synchronization
```

10. Biometric Enrolment Flow

Employee exists in Web Software

```
↓
Employee synced to device
↓
Supervisor approaches device
↓
Supervisor Authentication
■■■ Fingerprint
```

```
■■■ PIN
↓
Select Employee
↓
Start Enrolment
↓
Capture Finger #1
↓
Capture Finger #2
↓
Generate Template Locally
↓
Upload Template via HTTPS
↓
Enrolment Successful
```

• Employee unavailable

• Fingerprint quality too low

• Fingerprint enrolled successfully

```
UX states:
```

11. Device Registry Flow

Devices

```
↓
Device Registry
↓
Device List
■■■ Online
■■■ Offline
■■■ Warning
■■■ Pending Activation
```

• Device name

• Device ID

• Location

• Status

• Last heartbeat

• Firmware version

• Employee roster version

• Pending events

• Storage status

```
Device list:
```

12. Device Details Flow

Device Registry

```
↓
Select Device
↓
Device Details
```

```
Tabs:
Overview / Attendance / Roster / Biometrics / Logs / Actions
```

• Sync roster

• Restart

• Refresh status

• Send configuration

• Update firmware

```
Actions:
```

13. Device Provisioning Flow

Devices

```
↓
Add / Provision Device
↓
Enter Device ID
↓
Select Location
↓
Generate / Assign Credentials
↓
Verify Device
↓
Activate
↓
Device Online
```

```
Device states:
Pending → Provisioned → Connected → Active
```

14. Device Offline Flow

DEVICE OFFLINE

```
Branch: Abuja
Device: Terminal 04
```

```
Last seen: 10:42 AM
Attendance: 127 events stored locally
Status: Recording attendance offline
```

[View Device]

```
When connection returns:
Connection Restored
↓
Uploading buffered events
↓
127 / 127 uploaded
↓
Reconciliation Complete
↓
Device synchronized
```

15. Device Storage Warning Flow

Device Storage

```
↓
Approaching Capacity
↓
Alert Dashboard
↓
Admin Notification
↓
Device continues capturing
↓
Operator action required
```

16. Locations / Branches Flow

Locations

```
↓
Location Directory
↓
Select Location
↓
Location Details
```

• Location name

• Location reference

• Active employees

• Devices

• Online devices

• Offline devices

• Today's attendance

• Recent alerts

```
Location page:
```

```
Location drill-down:
Location
■■■ Attendance
■■■ Employees
■■■ Devices
■■■ Analytics
■■■ Alerts
```

17. Location-Scoped Access Flow

Login

```
↓
Identify Role
↓
Identify Assigned Locations
↓
Load Authorized Data
↓
Dashboard
```

• All locations

• All devices

• All employees

• System configuration

• Audit logs

```
Administrator:
```

• Assigned locations

• Attendance

• Employees

• Devices

• Reports

```
Manager:
```

• Assigned branch

• Attendance

• Employees

• Branch devices

• Limited device operations

```
Branch Supervisor:
```

18. Users & Roles Flow

Administration

```
↓
Users & Roles
↓
User Directory
↓
Select User
↓
User Profile
```

```
Add User:
Add User
↓
Enter User Information
↓
Select Role
↓
Assign Locations
↓
Configure MFA Requirement
↓
Save
```

• Administrator

• Manager

• Branch Supervisor

```
Roles:
```

19. Alerts Flow

Alerts

```
↓
Alert Center
↓
```

```
Filter
■■■ Critical
■■■ Warning
■■■ Info
■■■ Device
■■■ Sync
■■■ Integration
```

• What happened?

• Affected device/location

• Time

• Impact

• Recommended action

• Resolve / Acknowledge

```
Alert detail:
```

• Device offline

• Unmapped location

• Sync failed twice

• Storage approaching capacity

• Attendance delivery failure

• Firmware issue

• Integration failure

```
Examples:
```

20. Integration / Synchronization Flow

Administration

```
↓
Integration & Sync
↓
Integration Status
```

• Employee Sync — Healthy

• Location Sync — Healthy

• Attendance Push — Healthy

• Last Sync

```
Dashboard:
```

```
Sync history:
Sync History
↓
Select Sync Run
↓
Sync Details
```

• Start time

• End time

• Records processed

• Records added

• Records updated

• Records removed

• Errors

• Failed records

```
Display:
```

21. Sync Failure Flow

Sync

```
↓
Failure #1
↓
Retry
↓
Failure #2
↓
Create Alert
↓
Notify Administrator
↓
Admin opens alert
↓
View Sync Failure
↓
Investigate
↓
Retry Sync
```

22. Unmapped Location Flow

Employee Sync

```
↓
Employee location reference
↓
No matching platform location
↓
DO NOT discard employee
↓
Create Unmapped Location Alert
↓
Admin opens alert
↓
Map external location
↓
Re-run synchronization
↓
Employee assigned correctly
```

23. Audit Log Flow

Administration

```
↓
Audit Log
↓
Search / Filter
↓
```

```
Select Event
↓
Audit Detail
```

• Actor

• Role

• Action

• Timestamp

• IP/device information where available

• Object affected

• Before state

• After state

```
Audit event:
```

24. Firmware Management Flow

Devices

```
↓
Firmware
↓
Firmware Versions
↓
Select Version
↓
Select Devices
↓
Review Deployment
↓
Deploy
↓
Monitor Deployment
```

```
Deployment states:
Queued → Downloading → Installing → Restarting → Verified
```

```
Failure:
Installation failed
↓
Device remains protected
↓
Retry / Rollback
```

25. System Settings Flow

Administration

```
↓
Settings
```

• Security

• Authentication

• MFA

• Session timeout

• Password policy

• Notifications

• Integration

• Device policies

• Data retention

• Organization settings

```
Sections:
```

Every mutation must generate an audit entry.

26. Critical Global States

```
Loading:
Loading attendance data...
```

```
Empty:
No attendance records found.
```

```
Offline:
Connection unavailable.
Showing the latest synchronized data.
```

```
Error:
Something went wrong.
[Try Again]
```

```
Permission denied:
You don't have permission to access this resource.
```

```
Syncing:
Synchronizing...
```

```
Success:
Synchronization completed successfully.
```

```
Partial success:
Synchronization completed with 3 errors.
[View Errors]
```

27. Core Navigation Architecture

LOGO

Overview

ATTENDANCE

• Live Attendance

• Attendance History

• Reports & Analytics

WORKFORCE

• Employees

• Locations

DEVICES

• Device Registry

• Firmware

SYSTEM

• Alerts

• Integration & Sync

• Users & Roles

• Audit Log

• Settings

User Profile

Logout

28. Complete Page Inventory

1. Login

2. MFA Verification

3. Account Locked

4. Forgot Password

5. Reset Password

6. First-Time Setup

7. MFA Setup

8. Recovery Codes

```
Authentication:
```

9. Dashboard

10. Dashboard Empty State

11. Dashboard Error State

```
Dashboard:
```

12. Live Attendance

13. Attendance History

14. Attendance Detail

15. Reports & Analytics

16. Report Configuration

17. Report Results

```
Attendance:
```

18. Employee Directory

19. Employee Profile

20. Employee Attendance

21. Employee Biometric Status

22. Employee Sync Status

```
Employees:
```

23. Location Directory
24. Location Details
25. Location Attendance
26. Location Employees
27. Location Devices
28. Location Analytics

```
Locations:
```

29. Device Registry
30. Device Details
31. Device Attendance
32. Device Roster
33. Device Biometrics
34. Device Logs
35. Device Provisioning
36. Device Activation
37. Device Offline State
38. Device Storage Warning
39. Firmware Versions
40. Firmware Deployment
41. Firmware Deployment Details

```
Devices:
```

42. Users
43. User Details
44. Create User
45. Roles & Permissions
46. Alerts
47. Alert Details
48. Integration Dashboard
49. Sync History
50. Sync Details
51. Unmapped Location Resolution
52. Audit Log
53. Audit Detail
54. Settings
55. Security Settings

56. Notification Settings

57. Integration Settings

```
Administration:
```

58. 404

59. 403 Forbidden

60. 500 Error

61. Maintenance

62. Session Expired

```
System States:
```

29. Highest-Priority User Journeys

```
Journey A — Manager checks attendance
```

Login

```
↓
MFA
↓
Dashboard
↓
Live Attendance
↓
Filter Location
↓
Select Employee
↓
Employee Attendance
```

```
Journey B — Employee gets enrolled
```

Employee created in source web software

```
↓
Employee sync
↓
Employee appears on device
↓
Supervisor authentication
↓
Select employee
↓
Capture fingerprint
↓
Generate template
↓
Upload
↓
Success
```

```
Journey C — Device goes offline
```

Device online

```
↓
Internet lost
↓
Device switches offline
↓
Employee authenticates
↓
```

```
Attendance stored locally
↓
Dashboard shows device offline
↓
Internet restored
↓
Events upload automatically
↓
Reconciliation
↓
Dashboard updated
```

```
Journey D — Employee is terminated
```

Employee becomes inactive

```
↓
Web software sync
↓
Platform receives update
↓
Employee removed from device roster
↓
Biometric templates removed
↓
Historical attendance retained
```

```
Journey E — Integration failure
```

Sync

```
↓
Failure
↓
Failure again
↓
Alert created
↓
Admin opens alert
↓
Sync details
↓
Fix issue
↓
Retry
↓
Success
```

```
Journey F — Device management
```

Device Registry

```
↓
Device Details
↓
Check health
↓
Send remote command
↓
Monitor response
```

30. UX Principle Behind the Whole Product

The UI should communicate system health, not just display data.

```
Every important object should answer four questions immediately:
```

1. What is happening?

2. Is something wrong?

3. Where is it happening?

4. What can I do about it?

```
Example:
```

DEVICE OFFLINE

Lagos Branch · Terminal 04

```
Last seen: 10:42 AM
Buffered events: 127
Employees affected: 86
```

[View Device] [View Location]

rather than simply:

```
Device: Offline
```

31. Recommended Design Hierarchy

PHASE 1

Information Architecture

```
↓
PHASE 2
User Flows
↓
PHASE 3
Wireframes
↓
PHASE 4
Component System
↓
PHASE 5
High-Fidelity UI
↓
PHASE 6
Prototype
↓
PHASE 7
Usability Testing
```

Do not start with colors, cards, gradients and fancy dashboard widgets.

```
First lock down:
```

```
roles → permissions → navigation → states → workflows → exceptions → UI.
```

The PRD is fundamentally an operations and reliability product. Its UX therefore needs to make synchronization, device health, attendance delivery, offline operation and exceptions highly visible rather than hiding them behind attractive dashboard cards.